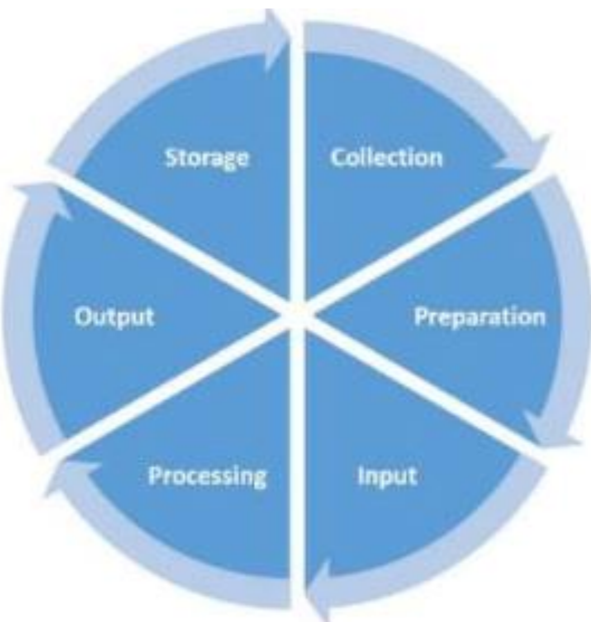
Data Processing Cycle

Data Processing Cycle

- The data processing cycle refers to a series of steps where raw data (input) is fed into a process to produce actionable insights (output).
- It includes six stages/phases where the raw data can be taken into in order to get the desired results. These stages can be summarized as shown from figure below:

WHAT IS THE PURPOSE OF DATA PROCESSING CYCLE?
ANSWER
it is just to process the raw data to produce the actionable insight



WHY CYCLE IN PROCESSING THE DATA
ANSWER
because the data management is dynamic , and each steps is essential for continuous improvement and relevance

due to dynamic nature in managing the data

Source: https://planningtank.com/computer-applications/data-processing-cycle

Phases of Data Processing Cycle

WHAT IS THE DIFFERENT BETWEEN THE PRIMARY DATA AND SECONDARY DATA

- **Data Collection:** The data can be collected from primary (field) or secondary source. It is a process where the data are collected and gathered from different sources-primary or secondary sources. This is the first step which provide data for the input.

  Primary data: tis is just the first hand data collected by resercher him self example interviews, observation etc whiile

  secondary data sources this is just the data that has already collected and processed by the others such as books

- **Data preparation**: Is the process of sorting and filtering the raw data to remove unnecessary and inaccurate data. It is often referred as "pre-processing" stage at which raw data is cleaned up and organized for the next stage of data processing

  why do we preparing the data: because in order to get the expected output

- **Data Input**: This is the feeding of raw and sieved (cleaned) data for processing. In this step, the raw data is converted into machine readable form and fed into the processing unit

- **Data Processing:** This is the step where data is processed by various methods such as electronic data processing, mechanical processing, or automated means. For example the input data can be processed using machine learning algorithms to generate desired output.

- **Data Output:** This is a stage where the processed data is finally transmitted and displayed to the user in a readable form, such as graphs, tables, audio, video, documents, etc.

- **Data Storage**: This is a stage where the data and metadata are stored for further use. This allows for quick access and retrieval of information whenever needed, and also allows it to be used as input in the next data processing cycle directly.

Big Data Processing Cycle

Question. (Group Work @ 10 marks)

What can you say about traditional data processing Cycle and Big Data Processing Cycle? Discuss

Big Data Architecture

Big Data Architecture:

- Big Data Architecture is a framework that lays out the blueprint of providing solutions and infrastructures to handle big data depending on an organization's needs. It is a comprehensive system of processing a vast amount of data.

- It is the framework that serves as a reference blueprint for big data infrastructures and solutions which logically defining how big data solutions will work, the components that will be used, layers to be used, and how information will flow.

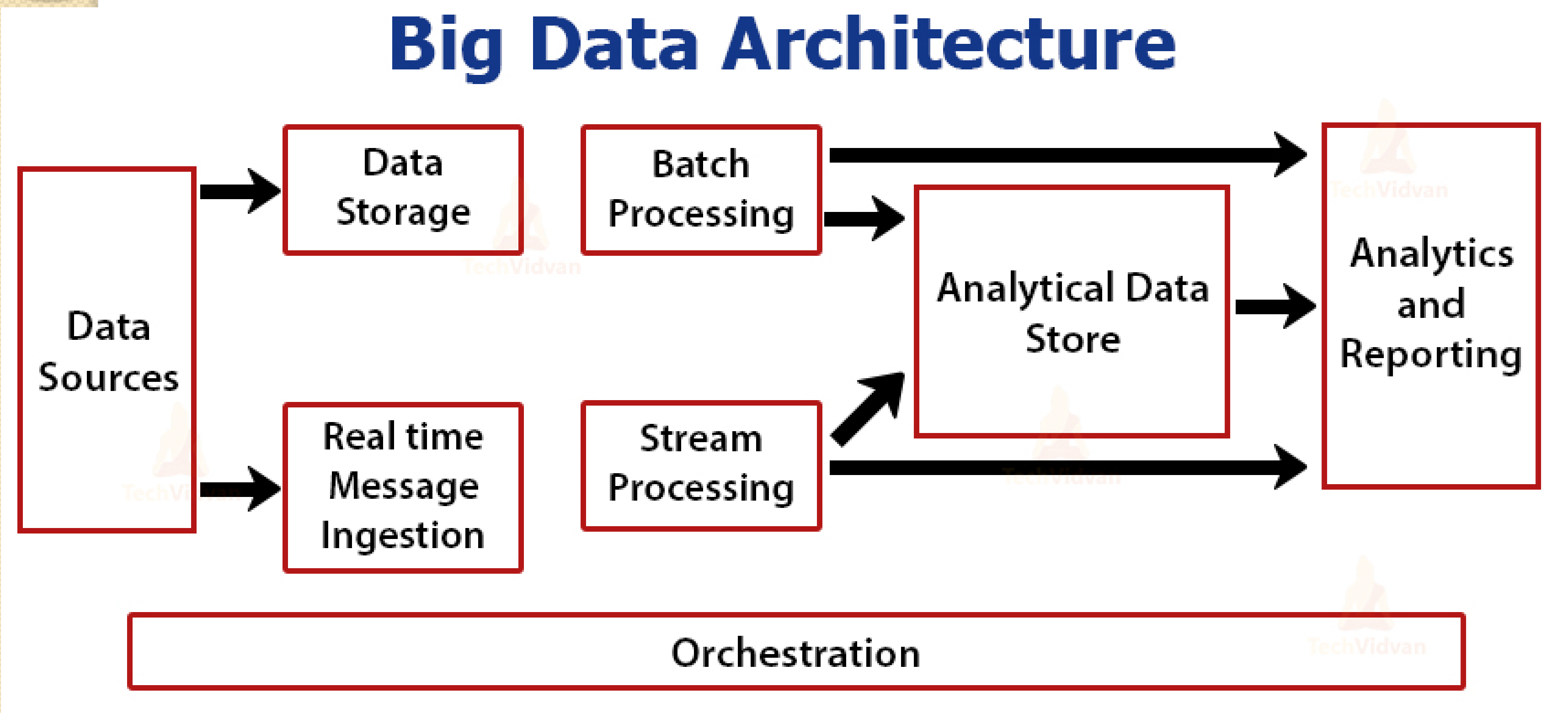
Big data solutions can involve one or more of the following workload:

- Batch processing of big data.
- Real-time processing of big data.
- Interactive exploration of big data.
- Predictive analytics and machine learning.

Big Data Architecture

Big Data Architecture:

- The architecture components of big data analytics typically consists of the following logical layers which preform the following major processes



Big Data Architecture

Components of Big Data Architecture :

- **Big Data sources**: Big data solution should have one or more data sources. The data sources are both open and third-party data providers. These include application data stores, such as relational databases, Static files produced by applications, such as web server log files. The data managed can be both batch processing and real-time processing
- **Data storage:** The storage layer is the second layer in the architecture receiving the data for the big data. It provides infrastructure for storing past data, converts it into the required format, and stores it in that format. For instance, the structured data is only stored in RDBMS while batch processing data is stored in the Hadoop Distributed File System (HDFS). This kind of store is often called a data lake
- **Batch processing and real-time processing:** The big data architecture needs to incorporate both types of data: batch (static) data and real-time or streaming data.
  - **Batch processing**: Because the data sets are so large, often a big data solution must process data files using long-running batch jobs to filter, aggregate, and otherwise prepare the data for analysis. The way the batch process works is to read the data into the storage layer, process it, and write the outputs into new files. The solution for batch time processing is Hadoop
  - **Real-time message ingestion**: The architecture must include a way to capture and store real-time messages for stream processing. Real-Time Processing is required for capturing, storing, and processing the data on a real-time basis. Real-time processing first ingests the data and then uses that as a "publish-subscribe kind of a tool."

Big Data Architecture

Components Big Data Architecture

- **Stream processing:** This processing varies from real-time message ingestion. Stream processing handles all the window streaming data or even streams and writes the streamed data to the output area. The tools here are Apache Spark, Apache Flink, Storm.
- **Analytical data store:** This is a one-stop place for the prepared data. This data is either presented in an interactive database that offers the metadata or in a NoSQL data warehouse. The data set then prepared can be searched for by querying and used for analysis with tools such as Hive, Spark SQL, Hbase.
- **Analysis and reporting:** The analysis layer interacts with the storage layer to extract valuable insights and business intelligence. The goal of most big data solutions is to provide insights into the data through analysis and reporting
- **Orchestration**: Most big data solutions consist of repeated data processing operations, to automate these workflows, you can use an orchestration technology such Azure Data Factory or Apache Oozie and Sqoop. Data orchestration is a process of coordinating and automating the data workflow in order to avoid get incorrect insights.

Big Data Architectural Layers

Big Data Architecture can be classified into four Layers**:**

- **Big Data Sources Layer**: a big data environment can manage both batch processing and real-time processing of big data sources, such as data warehouses, relational database management systems, SaaS applications, and IoT devices.
- **Storage Layer**: receives data from the source, converts the data into a format comprehensible for the data analytics tool, and stores the data according to its format.
- **Analysis Layer**: analytics tools extract business intelligence from the big data storage layer.
- **Consumption Layer:** receives results from the big data analysis layer and presents them to the pertinent output layer - also known as the business intelligence layer.

Big Data Solution Example

Consider the following data problem**:(This question should be discussion in the practical session)**

- Most of the organization use sales forces as their CRM application
  - Business users wants to see real-time data for their analysis stream processing
  - Business users want to take actions based on real-time data data analysis and visulization/ reporting
  - Business users want to integrate sales force data with other system to see the insights data oestration

Question: Which Big Data Architectural component will be used in solving these data solution?